Hyperglycemia

Primary Survey

Secondary Survey

Hyperglycemia
RBS >250 mg/dl
AND
Signs of Ketoacidosis or Shock

Administer Normal Saline
Adults: 500–1000 mL IV bolus
reassess and repeat as needed (up to 2000mL)
Pediatrics: 10–20 mL/kg IV bolus
reassess and repeat as needed (maximum 60 mL/kg).

EMT

12-Lead ECG
(watch for hyperkalemia and Arrhythmias)

Consider Tele EMS

Consider Red Flags
- Severe hyperglycemia (RBS >400 mg/dL) with signs suggestive of DKA or HHS (e.g., Kussmaul respirations, altered mental status, dehydration).
- Hypotension (SBP <90 mmHg) or clinical signs of shock (delayed capillary refill, cold extremities).
- Fever or clinical signs of infection (suggesting sepsis as a precipitating factor).
- Persistent symptoms despite initial EMS intervention.
- Recurrent vomiting or reduced urine output.

Signs of DKA:
- Fruity odor on breath
- Kussmaul Respirations
- Confusion/Altered Mental Status
- Dehydration
- Tachycardia / Shock

Patient Safety Considerations
- Monitor for signs of hypoglycemia if patient receives treatment prior to EMS arrival.
- Avoid aggressive fluid resuscitation in patients with renal or cardiac comorbidities.
- Do not delay transport to initiate full fluid replacement therapy.
- Escalate early in patients with altered mental status or suspected DKA/HHS.

Patient Care Goals

1. Identify and manage hyperglycemia to prevent progression to diabetic ketoacidosis (DKA) or hyperosmolar hyperglycemic state (HHS).
2. Correct dehydration and support perfusion.
3. Recognize and escalate care in critical hyperglycemia presentations.

Patient Inclusion & Exclusion Criteria

- Inclusion Criteria:
  - Patients with elevated blood glucose >250 mg/dL.
  - Symptoms of polyuria, polydipsia, vomiting, weakness, or altered mental status.
- Exclusion Criteria:
  - Hypoglycemia (RBS <70 mg/dL).

Key Patient Assessment Points

1. **Glucose Measurement**
   - Check RBS to confirm hyperglycemia (>250 mg/dL).
   - Repeat RBS after interventions to monitor response.
2. **History Specific to Glycemic Disorders**
   - Confirm history of diabetes (type 1 or type 2), insulin use, or oral hypoglycemic agents.
   - Inquire about recent medication compliance (e.g., missed insulin doses)
   - Identify precipitating factors (e.g., comorbidities, infection, stress, dietary changes).
3. **Symptoms of Hyperglycemia**
   - Note signs of dehydration (e.g., dry mucous membranes, thirst, hypotension, sunken eyes).
   - Assess for symptoms of diabetic ketoacidosis (DKA) or hyperosmolar hyperglycemic state (HHS), such as fruity breath, Kussmaul respirations, or severe lethargy.
   - Evaluate mental status (GCS or AVPU).
4. **Determine Triage Category**
   - Determine Triage Category as per Patient Triage Categories CPG.
     - Red Priority: Hyperglycemia >324mg/dl WITH symptoms of DKA/HHS, or signs of shock.

Patient Management & Interventions

1. **Primary Survey**
   - For unresponsive patients, perform airway assessment and initiate appropriate airway management.
   - Place unconscious patients in the recovery position at minimum while treatment is being initiated.
2. **Secondary Survey**
   - Complete a focused assessment, including full vital signs, glucose level, and mental status (GCS/AVPU).
   - Perform a 12-lead ECG to evaluate for signs of hyperkalemia (e.g., peaked T-waves, widened QRS) or acute coronary syndrome, especially in elderly or high-risk patients.
3. **Consider Red Flags**
   - Severe hyperglycemia (RBS >400 mg/dL) with signs suggestive of DKA or HHS (e.g., Kussmaul respirations, altered mental status, dehydration).
   - Hypotension (SBP <90 mmHg) or clinical signs of shock (delayed capillary refill, cold extremities).
   - Fever or clinical signs of infection (suggesting sepsis as a precipitating factor).
   - Persistent symptoms despite initial EMS intervention.
   - Recurrent vomiting or reduced urine output.
4. **Consider ALS back-up**
   - Consider the requirement for ALS back-up as per Universal Care CPG.
5. **Evaluate for Signs of Shock or Instability**
   - Hypotension, tachycardia, prolonged capillary refill, or altered sensorium.
   - If signs of dehydration, shock or DKA are present, initiate fluid resuscitation.
6. **Establish IV Access**

- Indicated in patients with hypotension, persistent altered mental status, or severe hyperglycemia.
- Consider intraosseous (IO) access if peripheral IV access is not obtainable.

7. **Fluid Resuscitation**
   - Administer Normal Saline or Ringer's Lactate for hyperglycemia with signs of dehydration, shock or DKA:
     - Adults: 500–1000 mL IV bolus (up to 2000mL), reassess and repeat as needed.
     - Pediatrics: 10–20 mL/kg IV bolus (up to 1000mL) reassess and repeat as needed (maximum 60 mL/kg).
   - Monitor for signs of fluid overload, especially in patients with cardiac or renal disease.
8. **Patient Reassessment**
   - Frequently Reassess the patient as per Universal Care CPG.
   - Specific Patient Reassessment Points:
     - **Glucose Monitoring**
       - Reassess blood glucose every 15 minutes after treatment or fluid administration.

Patient Safety Considerations

- Monitor for signs of hypoglycemia if patient receives treatment prior to EMS arrival.
- Avoid aggressive fluid resuscitation in patients with renal or cardiac comorbidities.
- Do not delay transport to initiate full fluid replacement therapy.
- Escalate early in patients with altered mental status or suspected DKA/HHS.

Clinical Insights & Additional Guidance

- Document suspected cause of hyperglycemia (infection, missed insulin, stress).
- Confirm recent medication compliance or omission (insulin or oral agents).
- Assess for abdominal pain or vomiting in DKA patients.
- Avoid oral fluids if patient has altered mental status.
- Inform receiving facility if patient meets criteria for DKA/HHS for early preparation.

Documentation

- Identification: Full patient name, age, and gender.
- Mechanism / Medical Complaint: Hyperglycemia symptoms such as weakness, nausea, vomiting, or decreased level of consciousness.
- Injuries / Information: Diabetes history, insulin omission, precipitating factors like infection.
- Signs: Vital signs, dehydration indicators, AVPU or GCS, Kussmaul breathing, RBS.
- Treatment:
  - IV fluid therapy: type, volume, rate.
  - Oxygen therapy if indicated.
  - Repeat glucose level if performed.
- Allergies: Relevant to medications or fluids.
- Medications: History of missed insulin or oral antidiabetics.
- Background: Comorbidities (renal, cardiac), prior DKA/HHS episodes.
- Other: Family input, home medications, previous EMS calls, hospital notification.

Key Performance Indicators

1. Administer IV fluid for patients with dehydration of signs of shock.

References

1. National Association of State EMS Officials (NASEMSO) (2022) *National Model EMS Clinical Guidelines*. Available at: https://nasemso.org/wp-content/uploads/National-Model-EMS-Clinical-Guidelines_2022.pdf
2. Joint Royal Colleges Ambulance Liaison Committee (JRCALC) (2006) *Glycaemic Emergencies in Adults*. Available at: https://warwick.ac.uk/fac/sci/med/research/hsri/emergencycare/prehospitalcare/jrcalcstakeholderwebsite/guidelines/glycaemic_emergencies_in_adults_2006.pdf (Accessed: 29 July 2025).

3. BC Emergency Health Services (BCEHS) (2024) *Clinical Practice Guidelines: E01 Hypoglycemia and Hyperglycemia*. BCEHS Handbook. Available at: https://handbook.bcehs.ca/clinical-practice-guidelines/e-medical-emergencies/e01-hypoglycemia-and-hyperglycemia/
4. St John New Zealand (2025) *Hyperglycaemia*. Clinical Procedures and Guidelines. Available at: https://cpg.stjohn.org.nz/tabs/guidelines/metabolic-poisoning-eas/page/hyperglycaemia-eas
5. Queensland Ambulance Service (QAS) (2025) *Clinical Practice Guidelines: Medical/Diabetic emergency: Hyperglycaemia*. Available at: https://www.ambulance.qld.gov.au/__data/assets/pdf_file/0022/219046/cpg_diabetic-emergency_hyperglycaemia.pdf (Accessed: 29 July 2025).
6. Umpierrez, G.E., Hellman, R., Korytkowski, M.T., Kosiborod, M., Maynard, G.A., Montori, V.M., Seley, J.J. and Van den Berghe, G. (2012) 'Management of hyperglycemia in hospitalized patients in non-critical care setting: an endocrine society clinical practice guideline', *The Journal of Clinical Endocrinology & Metabolism*, 97(1), pp.16-38.
7. Dhatariya, K.K. and Vellanki, P. (2021) 'Treatment of diabetic ketoacidosis (DKA)/hyperglycemic hyperosmolar state (HHS): novel advances in the management of hyperglycemic crises (UK versus USA)', *Current Diabetes Reports*, 17(5), p.14.